

Konrad Kleczkowski

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Skills

Languages: Haskell, Rust, C, C++, Java, Scala, Python, Bash

Infrastructure: Docker, AWS, Azure, Kubernetes

Productivity: Git, Perforce, Jira, Bitbucket, ADO

Experience

Quantitative Developer *Remote*
Standard Chartered Bank *June 2021 – Jan 2026*

- Designed and implemented a real-time xVA calculation platform, covering both system architecture and development of core components.
- Led the migration from an end-of-day (EOD) valuation process to a fully real-time computation model, significantly improving risk visibility and operational responsiveness.
- Collaborated directly with business stakeholders and risk teams to refine requirements, validate models, and ensure alignment with regulatory expectations.
- Delivered the production-ready system successfully in September 2025, achieving stable performance under high-throughput workloads.
- Built performance-critical components using **Haskell**, **Rust**, and **C++**, focusing on correctness, concurrency, and low-latency execution.

Blockchain Developer *Warsaw, Poland*
Blockhunters *Jan 2020 – May 2021*

- Developed a platform for automated validation and verification of Ethereum smart contracts, focusing on correctness, safety, and exploit detection.
- Designed and implemented a custom virtual machine capable of symbolic execution of contract fragments to uncover edge cases, vulnerabilities, and unexpected execution paths.
- Integrated the Z3 SAT/SMT solver to reason about contract behavior, detect logical inconsistencies, and identify potential security flaws.
- Built performance-critical components in **Rust**, **Haskell**, and **C++**, emphasizing reliability, formal rigor, and efficient symbolic evaluation.
- Collaborated with blockchain engineers and auditors to refine verification strategies and improve detection coverage across diverse contract patterns.

Compiler Engineer *Gdańsk, Poland*
Intel *Nov 2018 – Dec 2019*

- Developed and maintained the GLSL compiler for Intel Iris GPU architecture, ensuring full compliance with hardware specifications and shading language standards.
- Designed and modeled the compiler's intermediate representations (IR) and generated GPU-specific machine code aligned with Intel's graphics processor documentation.
- Built internal debugging and inspection tools to analyze generated code paths, execution behavior, and GPU-level performance characteristics.
- Worked on an LLVM-based compilation pipeline written in **C++** supporting both ahead-of-time (AOT) and just-in-time (JIT) modes.
- Optimized critical compilation stages using deep knowledge of CPU and GPU microarchitecture, achieving a 2.5 times reduction in compilation time.
- Collaborated with GPU architects and driver teams to refine code generation strategies and improve shader execution efficiency.

Software Engineer *Wrocław, Poland*
Bright IT *Jan 2018 – Oct 2018*

- Designed and implemented web solutions based on the Magnolia CMS platform, focusing on modularity, maintainability, and seamless content authoring workflows.
- Developed components and integrations primarily in Scala and Java, ensuring high reliability through automated testing and continuous integration practices.
- Built and maintained a library enabling cross-CMS component translation, allowing a single UI widget to be mapped to multiple CMS ecosystems.
- Created automated test suites validating component behavior across different CMS targets, improving consistency and reducing regression risk.
- Collaborated with designers and backend teams to deliver cohesive, production-ready features aligned with client requirements.

Education

Wrocław University of Science of Technology *Wrocław, Poland*
B.E. in Computer Science *Oct 2016 – Jan 2020*

- GPA: 5.14/5.5, Distinction
- Co-Founder of Functional Programming Student Club